

Submission Cover Sheet

Course Details

Title: Blockchain Fundamentals
Code: DAT5002 S1_25
Term: Autumn 2025
Assessment: **PRAC1 – Task 3.2 (Reflection)**

Student Details

Tom Rowan ST20285213

Project Tutor

Dr. Ali Shahaab

Table of Contents

<i>Course Details.....</i>	<i>1</i>
<i>Student Details.....</i>	<i>1</i>
<i>Project Tutor.....</i>	<i>1</i>
<i>Section 3 – Blockchain Development (Reflection).....</i>	<i>3</i>

Section 3 – Blockchain Development (Reflection)

Task 3.2 (Individual)

I work in cyber security as a pentester. I have worked both for consultancy firms and as an internal tester. Hence I have experience of both bidding for work and granting contracts to external firms to provide testing. In my team, we decided to model this process in our “Pentesting Marketplace” application.

There are three types of account in the application: the Owner, Customers and Pentesters. While we could utilise the OpenZeppelin “Ownable” contract, we needed to write custom functions and modifiers to implement further access control. Only appropriate individuals can raise Requests for Proposals (RFPs) and different accounts can bid on them.

We tried to be efficient with ‘gas’ expenditure. Functions make use of memory variables, and act as views wherever possible. We also avoided hardcoding design features such as the app usage fee, and have provided functions for the Owner to make changes after deployment.

It is important to note that in reality, the application would have a web front end where most of the presentation of data would be handled. We have tried to limit gas usage on the block-chain, following the design methodology that “the web app would handle this for free.”

My cyber security background lends itself to thinking about access control and limiting access to various parts of the application data. We implemented the majority of code as either private or internal functions, and all data-structures are similarly hidden from direct external access.

I believe that we have done a good job of designing and implementing an app to model a real-world bidding process. I found Solidity a very effective language to work in from a security standpoint, and found it rewarding to implement our designs.

I am good at code documentation, so I made sure that the code was commented and that the accompanying ReadMe files were in good order.

I find working in groups a challenge in the degree apprenticeship context because it is difficult to find times when everyone can get together to discuss or work in the same space. Some of the team was able to make good use of the collaboration features in GitHub such as Pull Requests and Issues to communicate the code we were working on and the changes that we'd made. Sadly not everyone was able to put in equal effort.

(385 words)